Fig: A Schelling-Point Token for the Figaro Coordination Ecosystem

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Abstract

We describe Fig, the coordination token that ships alongside the Figaro settlement primitive. Fig is designed as a *Schelling-point token* in the sense of Schelling [1960]: a focal point around which participants in the Figaro ecosystem can coordinate without central authority or explicit agreement. Its value as a coordination artifact derives from its focality, not from any on-chain utility function, yield stream, governance claim, or fee path.

This framing inverts the usual design question for a new token. Rather than asking "what does FIG *do*?" we ask "what would Fig have to *not* do in order to remain a credible focal point?" The answer produces a catalogue of design exclusions, each of which removes a class of holding motivation that would dilute the purity of Fig's signaling value. Fig is not governance, not yield-bearing, not a fee path, not bond collateral for the kernel, not settlement-emitted, and not founder-vested. Each negative preserves focality by removing a reason to hold Fig that is not "I am signaling alignment with the Figaro ecosystem."

The positive design content is modest and targeted: a fixed one-billion-unit cap enforced by two complementary supply-integrity mechanisms, a one-way deployer-renouncement latch that freezes the allocation plan at deployment, and a single staged-merkle airdrop that distributes 60% of supply to pre-committed cohorts at years 2, 5, and 9. We lay this design out in detail, reject an earlier settlement-anchored "Euler oscillation" design for the focality-destroying reasons it introduced, and close with an honest analysis of how Fig can lose its focal-point status and what the design does and does not defend against.

The paper sits within a lineage. Nakamoto [2008] demonstrated that money could be a protocol rather than an institution, with Btc as the focal Schelling point for a peer-to-peer cash system. Buterin [2014] demonstrated that contracts could be composable, with Eth as the focal point for programmable-settlement. The present work extends the lineage one step: *institutions* themselves become composable through the Figaro primitive (mechanism and implementation papers), and Fig is the focal point for participants who coordinate around the composition.

Keywords: cryptoeconomics, Schelling point, focal-point coordination, token design, coordination token, supply integrity, decoupled emission

1 Introduction

Fig is the native token that ships alongside the Figaro settlement primitive developed in the mechanism and implementation papers. This paper describes its design and defends a specific framing of what it is. The framing is the substantive contribution; the supply-integrity and distribution mechanics follow from it.

The framing: *Fig is a Schelling-point token for the Figaro ecosystem.* Its value to participants is not a claim on protocol revenue (there is no protocol fee), not a governance vote (the kernel has no governance), not a yield stream (no activity-coupled emission), not a utility token

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purchasable for on-chain services (there are no protocol-layer services that require it), and not bond collateral (the kernel is token-agnostic). Its value is its focality: in the sense of Schelling [1960], Fig is the focal point that Figaro-aligned participants converge on when they want to signal alignment without coordinating through a central authority.

A lineage note. The design pattern has precedent. Nakamoto [2008]'s Bitcoin is, in retrospect, the clearest example of a Schelling-point token at the money layer: its value as a store of value derives from holders converging on it as *the* answer to the question "what digital bearer asset are we coordinating around?" The proof-of-work mechanism secures Bitcoin's ledger, but it does not manufacture Bitcoin's Schelling-point status — that emerged from participant convergence under conditions in which Bitcoin's design made convergence viable (fixed supply, open participation, no issuer). Buterin [2014]'s Ethereum moved the logic up one layer: Eth became the focal point for programmable settlement, with composable contracts as the substrate. The present work moves it up again: with Figaro, institutions themselves are composable, and Fig is the focal point for aligned participation in that composition. Each layer's native token has the same structural role — a focal coordination anchor that does not require a central issuer, does not require a governance apparatus, and cannot be replicated by a general-purpose token, because the signal it provides is specifically about alignment with that particular ecosystem. Each layer's token is also underpinned by a different primary economic quantity — Bitcoin by the cost of producing the asset, Ethereum by the demand for executing transactions, Fig by the trade conducted through the substrate (Section 3). Of the three, Fig's underpinning sits closest to where actual economic value is created, because trade is where it is created.

Paper organization. Section 2 introduces Schelling-point tokens as a design category and locates Fig within it. Section 3 argues why the Figaro ecosystem is well-suited to a focal-point native token rather than a utility-coupled one. Section 4 articulates design rules that support focal-point emergence and resist focality-dilution. Section 5 develops supply integrity. Section 6 develops allocation. Section 7 develops the staged-merkle airdrop. Section 8 catalogues the design exclusions and ties each to the focality property it preserves. Section 9 records the rejection of an earlier settlement-anchored "Euler oscillation" design and deepens the rejection rationale in light of the Schelling-point framing. Section 10 treats bootstrap and attack surface. Section 11 retains the scope note on substrate-neutrality and distributional silhouette. Section 12 reports the verification surface. Section 13 concludes.

2 Schelling-Point Tokens as a Design Category

Schelling [1960] identified *focal points* (what later came to be called Schelling points) as solutions that parties tend to choose in the absence of explicit communication, because some feature of the solution makes it stand out as the natural coordination target. Schelling's canonical example: if two parties are told to meet in New York City without being able to communicate, a disproportionate fraction converge on Grand Central at noon. Neither the location nor the time is more rational than any other; what makes them focal is a shared cultural salience that each party correctly expects the other to recognize.

Crypto and Schelling points. The structure has been rediscovered repeatedly in crypto design. Nakamoto [2008] did not use the word "Schelling," but Bitcoin's emergence as a store of value followed the logic precisely: a fixed-supply, non-upgradable, issuer-less digital bearer asset with a peer-to-peer monetary intent. No mechanism guarantees Bitcoin's SoV status; it emerged because the design made focal convergence viable and participants converged. Buterin [2014] extended the logic up a layer: Eth became the focal settlement unit for programmable contracts, for the same reason — the design made convergence viable (gas-required, ubiquitous across contracts, issuer-less at the base layer). Similar focal patterns have emerged for subsidiary positions within crypto: Uniswap as the focal AMM, Aave as the focal lending protocol, Lido as the focal liquid-staking wrapper. In each case the token (Uni, Aave, Ldo) has some utility, but a substantial fraction of its holding behavior is Schelling convergence on "the token that represents alignment with this particular primitive."

Schelling tokens vs. utility tokens. A pure utility token is valued for the services a holder can purchase with it (gas, access, data). A pure governance token is valued for the votes it conveys. A pure yield-bearing token is valued for the cash flows it represents. A Schelling-point token is valued for something different: its *focality*, which is a coordination property, not a cash-flow property or a rights property. A participant holds a Schelling-point token because holding it makes them legible to other aligned participants and because the shared holding constitutes the coordination itself.

Pure Schelling-point tokens are rare because most token designs add utility or governance features to create additional demand drivers. Each such addition, however, introduces holders whose reason for holding is *not* focal alignment — governance speculators, yield farmers, utility arbitrageurs — and these holders' presence dilutes the focality signal. Bitcoin's focality is durable partly because Bitcoin has *remained* a pure Schelling point by refusing to add governance or yield or utility features beyond the base-layer payment rail. Each refused feature protects the signal's purity.

Fig's design category. Fig is designed to sit in the pure-Schelling-point category. The design is committed to refusing utility couplings, governance rights, yield streams, and fee paths. The rest of this paper develops what that refusal looks like in practice.

3 Why the Figaro Ecosystem Is Well-Suited to a Focal-Point Token

What underpins Fig's value. A Schelling-point token requires focal convergence, but focal convergence coordinates *which* token participants pick; it does not by itself create value. Each token in the lineage invoked above has a distinct economic underpinning, sitting at a distinct layer of the economic stack.

- **Bitcoin** is underpinned by the *cost of production*: proof-of-work consumes real energy, hardware, and electricity, putting a floor on rational issuance. The underpinning is at the factor-input layer.
- **Ethereum** is underpinned by *transaction demand*: every Ethereum transaction requires gas denominated in Eth, with EIP-1559 burn reinforcing the floor. The underpinning is at the transaction-execution layer.
- **Fig** is underpinned by *trade*: every bonded commitment in the Figaro ecosystem is a real transfer of value — goods for payment, services for payment, coordination for payment. The aggregate volume and quality of these transfers is the economic content of the ecosystem; Fig is the token denomination of alignment with the value-adding activity that produces them. The underpinning is at the trade layer — the layer at which actual economic value is created.

This places Fig not at a weaker position in the lineage than its predecessors but at a different one, attached to a different primary economic quantity. Bitcoin is underpinned by the cost of the infrastructure it runs on; Ethereum by the demand for executing transactions on its substrate; Fig by the value produced when the substrate is used to coordinate trade. In classical-economics terms: factor cost, transaction utility, value-added. The three answers sit at three different points in the economic chain; Fig's is the most proximate to end-value creation.

A consequence worth stating explicitly: Fig's value is fully endogenous to Figaro's trade activity. If little trade flows through the ecosystem, Fig is correspondingly worth little; if substantial trade flows, Fig's value rises with it. This coupling is direct, on-chain visible, and participant-auditable — the volume of bonded commitments is a public ledger fact — which makes Fig's value-formation more legible than the factor-cost or transaction-fee underpinnings of its predecessors, both of which require external accounting (electricity prices, fee-market depth) to interpret. The coupling is the right relation between a coordination token and the activity it coordinates: the token is worth the trade it identifies aligned participants in, and no more.

Why a general-purpose token cannot carry this signal. The Figaro kernel is token-agnostic. Any ERC-20 can serve as the bond and payment currency for a bonded commitment (mechanism paper). The kernel does not need a native token to function, and Fig does not interact with the kernel in any equilibrium-relevant way. Why, then, have a native token at all?

The answer is that multi-party coordination in an ecosystem without central authority requires a coordination signal that does not depend on any central authority, and no general-purpose token provides that signal. Participants bond in `USDC` when they want legal-system alignment; in a DAO governance token when they want community membership in that DAO; in `ETH` when they want to denominate in the base settlement unit; in a commodity-backed token when they want value anchoring. None of these signals *Figaro-ecosystem alignment specifically.* A general-purpose token carries many signals; what distinguishes them cannot be inferred from the token.

A focal-point token that signals Figaro alignment *only*, and whose design forbids any other reading of the holder's motivation, provides a coordination instrument the ecosystem does not otherwise have. This instrument is useful in two distinct settings:

Finding aligned counterparties. A participant designing a new Figaro assembly (a bonded workflow composed from protocol components) wants to find counterparties who understand the protocol, operate in its idiom, and will engage constructively with its equilibrium discipline. Searching for such counterparties by settlement history is possible (§5 of the institutional-economics paper) but expensive — one must read the chain. Searching by Fig holding is cheaper: a participant who holds Fig has declared alignment through the specific act of acquiring a token whose only plausible reason to acquire is alignment.

Signaling alignment in contested contexts. Participants who want to declare Figaro-aligned participation in adjacent debates — regulatory, academic, industry — benefit from a token whose holding is unambiguous. A founder who accepts Fig in lieu of other compensation signals more than a founder who accepts `USDC`. Not every participant needs this signal; those who do benefit from the existence of a focal coordination anchor that has no financial pretext.

These use cases are modest. We do not claim Fig is indispensable or that the ecosystem fails without it. We claim only that a focal-point token has a coordination function unavailable to general-purpose tokens, and that a cohort of participants will find the function useful.

4 Design Rules for a Focal-Point Token

Supporting focal-point emergence is not the same as manufacturing it. A designer cannot declare that a token is a Schelling point; focality emerges from participant convergence or it does not. What the designer can do is remove features that would make convergence implausible or fragile, and add features that make convergence stable once it starts.

We articulate four design rules.

Rule 1: Pure signaling. The token must carry *one* unambiguous reason to hold it. Holders should be alignment-signalers, not yield-chasers, governance-speculators, utility-arbitrageurs, or fee-path-optimizers. Features that would introduce non-alignment holders are forbidden. This is the most restrictive rule and generates most of FIG's catalogue of exclusions (Section 8).

Rule 2: Predictable supply. Focality is fragile under surprise inflation. A holder who believes supply is fixed may exit on learning it is not. FIG's supply schedule is committed at deployment and cannot be modified: 1B hard cap, staged airdrop with three immutable merkle roots and immutable calendar unlocks, one-way deployer-renouncement at the end of the deploy transaction. There are no surprise-inflation paths available to any actor after deployment.

Rule 3: Clean distribution. Focality is fragile under highly concentrated ownership. If a single holder can dump enough supply to destabilize the price, Schelling convergence is less credible. FIG's allocation is 60% to a pre-committed community cohort (the staged airdrop), 30% to a treasury with an ecosystem mandate (the DAO), and 10% to founders. The concentration is higher than ideal for focality (the DAO + founder share is 40%), which we address honestly in Section 10 and in the distributional-silhouette concession in Section 11.

Rule 4: Absence of protocol-activity coupling. Focality requires that participants hold for alignment reasons. If the token's supply is a function of protocol activity, holders acquire the token by doing things that are not alignment (gaming the emission curve, wash-trading for rewards), and the signal is muddied. Activity-coupled emission is ruled out. We develop the argument at length in Section 9, where we describe an earlier FIG design that used settlement-anchored emission and explain why we rejected it on focality grounds.

5 Supply Integrity

FIG has a fixed maximum supply of 10^9 units (one billion, denominated in 18-decimal *ether* units in the contract). The supply ceiling is enforced by two distinct mechanisms operating at different times.

Per-mint enforcement (run-time). The `mint` function reverts with `SupplyCapExceeded` if `totalSupply()` + `amount` > `MAX_SUPPLY`, checked atomically inside a non-reentrant call.

Pre-registration enforcement (deploy-time). The contract maintains an aggregate `totalRegisteredCap`: the sum of all registered minter caps. `registerMinter` reverts if `totalRegisteredCap` + `cap` > `MAX_SUPPLY`. This means the entire allocation plan is committed to the 10^9 ceiling *before any token is minted.*

The pair (per-mint + pre-registration) is stronger than either alone in the focality frame: per-mint enforcement catches cap overruns that survive bugs; pre-registration enforcement puts the allocation plan on chain where a focality-assessing participant can read it directly rather than relying on off-chain narrative.

The renouncement latch. FIG has a one-way Boolean `deployerMintRenounced`, settable exactly once by the original deployer via `renounceDeployerMint`. After renouncement, `registerMinter` reverts and any deployer call to `mint` reverts. The deploy script (`script/DeployMainnet.s.sol`) registers exactly two minters — the deployer wallet (cap 4×10^8, used for the genesis mint) and the airdrop contract (cap 6×10^8) — and then calls `renounceDeployerMint` as the final action of the deploy transaction. After the transaction completes:

- `totalRegisteredCap` $= 10^9$ exactly;
- the deployer has minted exactly 4×10^8 to the designated founder and DAO wallets;
- only the airdrop contract retains residual mint authority, capped at 6×10^8;
- no path exists to register additional minters.

The supply schedule is therefore frozen by a single transaction, and the contract has no further "governance moments" available to anyone — not to the deployer, not to the DAO, not to any future upgrade proposal. In focality terms: a participant assessing Fig at time $t > t_{\text{deploy}}$ can verify on chain that no further minting will occur beyond the airdrop schedule. Focal convergence is not destabilized by uncertainty about supply.

6 Allocation: 10 / 30 / 60

The total supply is split:

Bucket	Share	Realization
Founder	10% (100M)	Genesis mint, no vesting, no unlock
DAO	30% (300M)	Genesis mint, no vesting, no unlock
Airdrop	60% (600M)	Three-stage merkle, yr 2 / 5 / 9

Founder allocation (10%). A modest founder share, by contemporary norms. The non-vesting choice is deliberate: a vesting cliff is a tacit promise of future voting alignment or schedule-driven holding, both of which would be reasons to hold Fig other than alignment signaling. Releasing the founder allocation at genesis exposes the founder to whatever market consequences attach to early sale; the structural observation is that an unvested founder forgoes an asymmetric option (sell into early demand if it materializes; ride out the vesting if it does not) that a vested founder retains, and we have chosen to forgo the option. The prior question — whether founder allocation at this magnitude is the right allocation at all — is a political-economy question we do not attempt to answer in a cryptoeconomic paper.

DAO allocation (30%). A treasury share, available immediately at genesis. The DAO is not a kernel-governance body — there is no kernel governance to be done — but a coordination treasury for ecosystem activity (grants, public goods, reference implementations). Releasing the DAO allocation at genesis means the treasury is operational from day one. A fair objection: doesn't a liquid 30% treasury with an ecosystem-grants mandate create exactly the reflexivity that Section 9 rejects in the Euler design, and doesn't it dilute Fig's focal-point purity (Rule 1)? We concede the objection in its general form. The defense is narrower than the objection implies: the coupling the Euler design introduces is *mechanism-level* (a deterministic on-chain function from settlement count to mint), while the coupling a treasury introduces is *policy-level* (a human-discretionary function from treasury governance to disbursement). Mechanism-level coupling is what the kernel's no-escape-hatches design principle rules out; policy-level coupling is attenuated by two layers of friction (off-chain discretion, revocable grants). The distinction is principled but not absolute: a sufficiently mechanical grant policy would reintroduce the coupling, which is why the DAO's operating procedures are themselves a design artifact that warrants its own treatment rather than a silent assumption.

Airdrop allocation (60%). The majority share, allocated over a calendar window with declining per-stage size. The structure is designed to (a) reach non-genesis participants, (b) reward sustained participation (longer-window cohorts at year 5 and year 9), and (c) avoid the "dump-on-launch" dynamic of single-event airdrops by spreading the supply emission over a 9-year horizon.

The 60/30/10 internal split of the airdrop (300M at yr 2, 200M at yr 5, 100M at yr 9) is front-loaded for the bootstrap reason: the network needs participants more in its early years than in its mature years. The declining schedule encodes that asymmetry into the supply.

A note on the specific percentages. The 10/30/60 split and the internal 50/33/17 staging of the airdrop reflect a design judgment rather than a first-principles derivation. Any front-loaded schedule (45/35/20, 40/40/20, 50/30/20) would satisfy the bootstrap argument above, and the specific weights are a calibration we do not attempt to justify beyond "plausible under the stated principles."

7 Cohort-Based Vesting via Staged Merkle

The 600M airdrop is realized by a single contract, `StagedMerkleAirdrop.sol`, with three immutable stages (`STAGE_COUNT` = 3). Each stage is parameterized by a `Stage` struct holding a merkle `root` and a `unlockTime`. Both are written once in the constructor and never mutated thereafter; there is no admin-controlled stage addition, schedule extension, or cohort modification.

Single contract, three stages. An earlier candidate design used three separate airdrop contracts. The single-contract design is preferred for two reasons. First, surface area: a single contract is a single audit target, a single set of formal-verification rules, and a single storage account. Second, atomic deploy guarantees: the `registerMinter` call binds the airdrop contract's 6×10^8 cap before the renouncement, and a single contract makes the cap binding atomic.

Leaf format. The merkle leaf is `keccak256(abi.encodePacked(account, amount))`, the canonical OpenZeppelin format. Choosing a non-canonical leaf format would have required custom tooling per off-chain proof generator and a corresponding audit cost; we accept the canonical choice.

One-shot per (stage, address). The `claimed[stageIndex][msg.sender]` double-mapping is set true on successful claim and checked on every entry. A given account can claim at most once per stage. No top-up, no re-claim, no rebate.

Calendar-based unlock. The unlock timestamps are calendar-based (Unix epoch seconds), not activity-based. Tying unlocks to settlement activity would re-introduce a coupling between protocol use and token availability — the category Rule 4 forbids. Calendar timing has the additional virtue of being predictable to all parties at deploy time, which supports focal-point clarity.

8 Design Exclusions as Focality Preservers

The cryptoeconomic content of Fig is largely defined by what the design excludes. We catalogue the exclusions and tie each explicitly to the focality property it preserves. This restates the exclusions documented in earlier drafts of the paper, but with the Schelling-point framing the reasons become unified: each exclusion removes a class of holder whose reason for holding FIG would be something other than alignment signaling.

FIG is not a governance token. The kernel has no governance surface to vote on. Adding a governance role to FIG would introduce governance speculators — participants who hold FIG because they want to vote on proposals, not because they want to signal alignment. This would dilute the signal by populating the holder set with non-alignment holders. Refusing governance preserves the signal's purity (Rule 1).

FIG is not yield-bearing. The kernel takes no protocol fee; there is no fee pool to distribute. A stake-and-earn mechanism would introduce yield farmers — participants who hold FIG for expected cash flow, not for alignment. Refusing yield preserves the signal's purity (Rule 1) and also preserves the activity-decoupling (Rule 4), since yield mechanisms are typically indirectly coupled to protocol activity.

FIG is not a fee path. Even if a future runtime layer above the kernel introduces a fee mechanism, that fee is not denominated in FIG. A fee-denomination role would introduce utility arbitrageurs — participants who acquire FIG to pay fees, not to signal alignment. The signal would be muddied by transactional holders.

FIG is not bond collateral for the kernel. The kernel's bonds are denominated in whatever ERC-20 the parties choose. Making FIG the preferred bond denomination would introduce collateral demand that distorts price discovery and couples FIG to kernel activity through the back door. Refusing this role keeps the kernel token-agnostic (per Paper A's design) and preserves FIG's focal-signal purity.

FIG is not founder-vested. The 10% founder allocation and the 30% DAO allocation are unvested. There is no cliff, no schedule, no time-locked vesting contract. Vesting would introduce schedule-driven holders — participants who hold FIG because they will receive more in the future, not because they are signaling current alignment. The vested-founder case is the clearest instance: a founder vesting into liquidity over four years holds FIG during that period for mixed reasons, only one of which is alignment.

FIG is not settlement-emitted. There is no path from kernel activity (commit, resolve, attest) to FIG mint. `FigaroBatchVerifier` — the layer-2 batch verification contract — is explicitly not a registered minter and will never be. The deploy script does not register it; the renouncement latch ensures it cannot be added later. This refusal is the most rigorous application of Rule 4 and is worth its own section (Section 9).

FIG is not a coordinator-pattern extension. The companion implementation paper (§7) develops a *coordinator pattern*: four sufficient conditions under which a mechanism composed with the kernel preserves the bonding equilibrium. FIG satisfies these conditions trivially: it has no kernel interaction at all. FIG and the kernel are siblings, not a composition.

9 Rejected Design: Euler-Oscillation Emission

An earlier iteration (preprint draft, February 2026) included a settlement-anchored emission mechanism we called *Euler oscillation*: a settlement-count-decayed cosine modulation on top of a halving envelope, with seller-only emission at resolution time. Three properties motivated it:

persistent bootstrap subsidy, partial wash-trade resistance, and seasonal coordination through a 1.5× peak rate.

We rejected it. The rejection deepens under the Schelling-point frame. In the prior draft we listed three reasons: reflexivity, partial-but-not-eliminative wash-trade defense, and oscillation-pattern speculation. The Schelling-point frame reveals a more fundamental reason: activity-coupled emission *destroys focal-point status* by introducing holders whose reason for acquiring FIG is not alignment.

The focality argument, formally. Under Euler emission at rate $r(n)$ with settlement-count decay and seller-only payout, a participant who settles a process through the kernel acquires FIG. The acquisition is triggered by the settlement event, not by a decision to signal alignment. After acquisition, the holder has FIG in their wallet — but the token-in-wallet does not distinguish "holder of FIG by alignment choice" from "holder of FIG because an Euler function happened to pay me." In the Schelling-point frame, these are two distinct populations whose signals interfere. A third party looking at the distribution of FIG holders cannot read focal alignment from the distribution because the distribution mixes two holder populations.

This is distinct from the reflexivity argument (which points at a decision-coupling between settlement choice and emission expectation) and the wash-trade argument (which points at a gameable payoff structure). It is a more fundamental point about what a token *means* under each design. Activity-coupled emission makes the token mean two things at once; pure airdrop distribution makes it mean one thing.

The wash-trade inequality, for completeness. Let the prevailing FIG/currency exchange rate be ϕ, let the seller-only emission rate at settlement count n be $r(n)$, let gas per cycle be g, and let the process duration be T with opportunity cost of capital ρ per unit time. A wash-trader bonds both sides: buyer posts $2P$, seller posts $2P$, total locked $4P$ for T, seller receives $r(n)$ FIG at resolution. The cycle is profitable iff

$$\phi \cdot r(n) \; > \; \rho \cdot 4P \cdot T \; + \; g. \tag{1}$$

Seller-only emission doubles the capital requirement from $2P$ to $4P$ but does not drive (1) to infeasibility. The parameter region where wash-trading is infeasible is non-empty but not universal. This was a secondary reason to reject Euler; the primary reason was the focality-destroying mixing of holder motivations above.

What we kept. The 600M airdrop bucket, the staged structure (now realized via calendar unlocks rather than settlement counts), and the design intent of distributing the bootstrap subsidy over time. The underlying intuition — that distribution should be staged rather than instantaneous — survived; the Euler mathematics did not.

10 Bootstrap and Attack Surface

A Schelling-point token cannot be declared; it must be *recognized*. This section is the most uncertain part of the design, because it concerns processes we cannot directly control.

Bootstrap: how FIG achieves focal status. Focal-point emergence for a newly-launched token depends on several converging factors, none of which the designer can guarantee.

- *Initial cohort.* The year-2 airdrop distributes 300M to a pre-committed cohort determined by a merkle root set at deploy. The cohort selection process (which is policy, not mechanism) determines who receives FIG in the first distribution and therefore who has the first opportunity to recognize FIG as a focal point.

- *Design legibility.* The design rules of Section 4, and the exclusions of Section 8, must be legible to prospective holders as consistent with focal-point coordination rather than as random design preferences. This paper is part of that legibility work.
- *Absence of competing focal candidates.* If an alternative token launched with better focal properties (cleaner distribution, more restrictive exclusions) at a moment when FIG had not yet stabilized, participants might converge on it instead. The design attempts to make FIG hard to beat on focal grounds (Rule 1 purity, Rule 2 supply predictability) but cannot preempt all competitors.

Focal-shift attack. The relevant attack against a Schelling-point token is the *focal-shift attack*: an adversary, through concerted action, moves focal convergence to a different token. The canonical example is the decade-long debate over whether Bitcoin-Cash or Bitcoin-SV could displace Bitcoin as the SoV focal point; neither succeeded, but both made the attempt at significant scale. The defense is Schelling stickiness: participants who have converged on a focal point typically hold through attempts to move the focus, because moving costs coordination value.

Dump attacks. An adversary with large FIG holdings can attempt to destabilize focal-point status by dumping supply. In a pure Schelling-point frame, this is *less damaging* than in a yield-anchored or governance-anchored frame: holders of FIG are holders by alignment choice, not by financial hedging, and the financial price of FIG is somewhat decoupled from its focal-point status. A price crash that leaves participant alignment unchanged does not necessarily destroy the focal point; a coordinated loss of participant alignment does.

Acquiescence attacks. The quietest attack is no attack at all: nobody converges, and FIG fails to become a focal point. This is possible and unavoidable. The defense is limited — we can support focal emergence but cannot force it. The honest statement is that if no cohort recognizes FIG as a focal point, the token has no Schelling-point value and operates as an unvalued distribution artifact. The supply-integrity and allocation properties continue to hold in that case; only the signaling function fails.

The griefing attack reconsidered. In earlier design iterations, a concern was raised about *griefing* — an adversary who writes code to drain the reward pool not for profit but for spite, given that no governance or fee-extraction capture is available. Under Euler, this attack had a concrete surface (manipulate the emission function to empty the reward budget). Under the current pure-airdrop design, the attack surface is narrower: the airdrop contract's claim path requires merkle-proof inclusion at a pre-committed address, and no drain is possible beyond what the merkle root already authorizes. The dump-at-unlock surface remains (addresses can claim at unlock and sell), but that is indistinguishable from the normal operation of the airdrop; it is not an exploit.

11 Substrate-Neutrality and Distributional Silhouette

The mechanism paper and the political-economy companion paper argue that the bonded settlement *kernel* is a substrate on which multiple organizational forms can be expressed. The present paper describes FIG, a specific token ecosystem that ships alongside the kernel. Those two statements are about different objects and a careful reader should not confuse them.

FIG's 10/30/60 allocation is *one* instantiation of a token economy over the substrate. The substrate is substrate-neutral; FIG is not. If the distributional silhouette of FIG — early founder allocation, large standing treasury, community recipients on a calendar schedule — rhymes with the silhouette of prior platform-capital arrangements that the political-economy paper critiques, that rhyme is a feature of this specific token's distribution, not of the substrate, and not inherent to the Schelling-point framing. Different focal-point tokens could be launched with different silhouettes (flatter, community-only, non-founder), and the Schelling-point design principles would apply to them unchanged.

We flag this explicitly to disentangle three claims:

- **Substrate-neutrality (true of the kernel)**: the bonded settlement primitive does not impose any particular distributional structure on tokens or organizations.
- **Focal-point integrity (true of FIG)**: the specific supply mechanism and exclusion catalogue enforce the allocation and the focality-preservation properties at deploy.
- **Allocation legitimacy (out-of-scope)**: whether 10/30/60 is the *right* allocation, whether a 30% liquid treasury is an appropriate ecosystem-coordination vehicle, and whether the implicit authority of the DAO over grant direction reproduces soft-governance patterns the political-economy companion critiques — these are political-economy questions the present paper does not resolve.

12 Verification Surface

The FIG ecosystem is verified independently of the kernel. We report rule and invariant counts as of the snapshot date; numbers will grow as the surface expands.

TLA$^+$ (`formal/FigToken.tla`). 8 invariants verified across 160,844 distinct states in approximately 9 seconds (TLC, breadth-first search, parameters `MAX_SUPPLY` = 5, 3 minters, 6 recipients): `Inv_MaxSupply`, `Inv_NonNegative`, `Inv_DeployerCannotMintAfterRenounce`, `Inv_NoMintToZero`, `Inv_MinterCap`, `Inv_BalancesSumToSupply`, `Inv_CapBelowMaxSupply`, `Inv_SupplyEqualsSumMinted`.

Halmos (`test/HalmosStagedMerkleAirdrop.t.sol`). 4 symbolic-execution proofs of staged-airdrop properties: claim flag set on success, one-shot enforcement per (stage, address), correct balance arithmetic, and merkle leaf format `keccak256(abi.encodePacked(account, amount))`.

Certora. 9 CVL rules across two specs:

- `certora/FigToken.spec` (6 rules): `totalSupplyWithinMaxSupply`, `totalRegisteredCapWithinMaxSuppl` `minterMintedWithinCap`, `totalRegisteredCapMonotonic`, `deployerMintRenouncedIsOneWayLatch`, `minterCapImmutable`.
- `certora/StagedMerkleAirdrop.spec` (3 rules): `claimedIsMonotonic`, `stageConfigImmutable`, `minterImmutable`.

Foundry regression. `test/fig/FigToken.t.sol` and `test/fig/StagedMerkleAirdrop.t.sol` cover the deploy flow, every custom-error revert path, and the canonical 10/30/60 allocation realized by `script/DeployMainnet.s.sol`.

Headline supply guarantee. The verification surface yields a strong guarantee: from the moment `deployerMintRenounced = true` is included in a finalized transaction, the `totalSupply()` trajectory is bounded above by 10^9, no new minters can be registered, and the only remaining mint authority is the airdrop contract whose schedule is itself frozen by its three immutable stages. The supply is deterministic conditional on the merkle leaves; the merkle leaves are committed at deploy. Focal-point participants assessing FIG can verify this directly from on-chain state.

13 Conclusion

Fig is a small design. Most of its positive content is procedural (supply cap, renouncement latch, staged merkle airdrop) and most of its substantive content is negative (no governance, no yield, no fee path, no founder vesting, no settlement-anchored emission, no coordinator-pattern role). Under the framing this paper develops, the negatives are not restraint without purpose — they are the specific design discipline required to support focal-point emergence. Every exclusion removes a class of holder whose presence would dilute the signaling value of Fig for participants who hold for alignment reasons.

The lineage, made explicit. Nakamoto [2008] took money out of institutions and made it a protocol. Bitcoin is the Schelling point at the money layer, underpinned by the cost of producing the asset: a bearer asset around which participants converge because its design makes convergence credible (fixed supply, issuer-less, peer-to-peer). Buterin [2014] made contracts composable and put Eth at the focal point of programmable settlement, underpinned by transaction demand: gas is the utility floor, Schelling recognition the coordination premium. The present work extends the pattern to institutions: bonded commitments are composable into arbitrarily large assemblies (mechanism and implementation papers), and Fig is the focal point for aligned participation in that composition, underpinned by *trade* — the value-added produced when parties exchange goods, services, or coordination through the bonded primitive. The pattern is the same at each layer: a native token that is not a utility, not a governance claim, and not a cash-flow instrument, but a coordination anchor whose value rests on a distinct real-economic quantity. The distinction between the three is which quantity: factor cost for Bitcoin, transaction utility for Ethereum, and the value-added of trade for Fig.

Open questions. Three questions remain outside this paper's scope but worth flagging. First, the airdrop cohort selection (which addresses appear in which stage's merkle root) is policy, not mechanism, and the policy framework should be developed and audited as a separate artifact before each stage's root is committed at deploy. Second, the relationship between Fig and the various treasury communities likely to form around it (cf. the "capital firm" discussion in the institutional-economics paper) is genuinely emergent and not centrally designable; the present paper does not attempt to predict or control that emergence. Third, the status of the specific percentages in the allocation is as addressed in Section 6: judgment, not proof; the structural choices are the defensible content.

What the design can and cannot do. Schelling-point design is a matter of supporting emergence, not causing it. The design rules of Section 4 and the exclusions of Section 8 make focal-point convergence on Fig viable: a participant who wants to signal alignment can do so by acquiring and holding Fig, and the design ensures the signal is not muddied by non-alignment holders. Whether focal-point convergence *happens* is a social process we do not control. We document the design so that those who choose to converge can do so legibly, and so that those who do not can see what they are declining.

Acknowledgments

The decision to frame Fig as a Schelling-point token, rather than as a decoupled-emission token whose cryptoeconomic content is largely negative, emerged in discussions with collaborators who pressed the question of what Fig *is* rather than what it *is not*. Their skepticism clarified the argument. The verification surface owes a continuing debt to the maintainers of TLA$^+$, Halmos, and Certora.

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